

NGUYEN TAT NGOC

Full-Stack Software Engineer | Node.js, NestJS, TypeScript, Laravel

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with over one year of experience building and deploying production web applications using TypeScript and PHP. Experienced in payment gateways, third-party API integrations, asynchronous job processing, RESTful API design, database optimization, and Docker-based deployment.

CORE TECHNICAL SKILLS

Backend & Architecture: Node.js, NestJS, TypeScript, PHP, Laravel, RESTful APIs, BullMQ, Background Job Processing.
Frontend: React, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Responsive Design.
Database & Cache: PostgreSQL, MySQL, Redis (Caching & Task Queue), Prisma ORM, Query Optimization & Indexing.
DevOps & Real-time: Docker, Docker Compose, Linux VPS (Nginx, Caddy Reverse Proxy, SSL), Git, Pusher, WebSockets.
Domain & Security: Payment Gateways (VNPay, ZaloPay), Logistics (ViettelPost API), AES-256-GCM, SSRF Mitigation.

WORK EXPERIENCE

NOVA AI SOLUTIONS CO., LTD — [Full-Stack Developer](#) Jan 2026 – Present

- E-Commerce & Logistics:** Built backend order workflows using Laravel/MySQL; integrated **ViettelPost API** to automate shipping rate calculations, waybill creation, and tracking, streamlining fulfillment.
- Payment Gateways & Security:** Integrated **VNPay** and **ZaloPay** APIs, preventing duplicate payment processing through webhook validation and idempotency handling; secured sensitive assets using private AWS S3 buckets with pre-signed URLs.
- QR-Order & Invoicing:** Developed multi-tenant table-ordering features; integrated **VNPT E-Invoice API** (SOAP/XML) for automated tax invoice generation and **Pusher WebSockets** for real-time alerts.
- Performance & Deployment:** Resolved N+1 queries through database indexing and eager loading, improving key API response times by ~35%; containerized and deployed services on Linux VPS using Docker and Nginx.

QUOC BAO SOFTWARE CO., LTD — [Mobile Developer Intern](#) Jun 2025 – Sep 2025

- Developed cross-platform mobile user interfaces using **Flutter** and **Dart**; managed application state with **GetX** and integrated backend RESTful APIs for authentication and dynamic content display.

FEATURED PROJECTS

UPLY – MULTI-TENANT WEBSITE & API MONITORING SAAS PLATFORM 2026

Role: Creator & Solo Developer • [Live Demo](#) • [GitHub](#)

- Decoupled Architecture:** Built a modular monorepo using **Turborepo**, comprising a **Next.js** dashboard frontend, a **NestJS** REST API backend, and an isolated background worker service.
- Distributed Job Queue:** Implemented task scheduling with **BullMQ** and **Redis** to execute recurring HTTP health checks every 60 seconds, detecting downtime and dispatching webhook alerts within seconds.
- Security & Multi-Tenancy:** Applied **AES-256-GCM encryption** for stored endpoint secrets, built an IP-filtering utility for **SSRF mitigation**, and enforced tenant-scoped data isolation through authorization checks and filtered Prisma queries.
- Deployment & Infrastructure:** Deployed frontend to Vercel and containerized backend/worker services with **Docker Compose**, using **Caddy** as reverse proxy with automated SSL certificates.

STUDENT CONDUCT SCORE & ACTIVITY MANAGEMENT SYSTEM (GRADUATION THESIS) 2025

Role: Lead Full-Stack Developer • **Defense Score:** 8.8 / 10 (Faculty Finalist) • [GitHub](#)

- Activity Recommender Engine:** Built an extracurricular activity recommender system using **K-Means Clustering**; normalized student profile vectors (One-Hot Encoding, Min-Max Normalization) to handle cold-start profiles.
- Anti-Fraud Attendance & Full-Stack:** Developed a time-windowed **QR Code check-in/check-out** mechanism to prevent attendance fraud; delivered **Laravel RESTful APIs** with 4-tier RBAC integrated with a **Flutter** mobile app.

EDUCATION

HO CHI MINH CITY UNIVERSITY OF INDUSTRY AND TRADE (HUIT) 2022 – 2026

Bachelor of Software Engineering • **GPA:** 3.37 / 4.00 • Classification: Very Good

Academic Merit Scholarship — Fall 2024